

Sentinel

Compliance Engineering Copilot

Built with **IBM Bob IDE**.

Turns HIPAA, SOC 2, and PCI from a periodic audit into a **daily-work habit**.

github.com/vaatus/sentinel · IBM Bob Hackathon · May 2026

The problem

Fortune 500 healthcare and fintech spend **\$5M–\$50M per company per year** on compliance audits.

Engineers chase PHI through codebases by hand for **6 months at a time** — and the violation gets shipped anyway.

Compliance is *periodic*. A quarterly audit. A release-gate scan. By the time a finding hits a Jira ticket, the engineer who wrote the violating line has moved on three sprints. The feedback loop is broken.

The insight

Compliance is a **whole-repo reasoning** problem.

A `logger.info(patient)` is only a HIPAA violation when seen through the route handler three files upstream that loaded the PHI.

- A regex sees the log line.
- A file-by-file AI sees the log line.
- An auditor's mental model sees the **flow** — and that's what `phi-tracer` mode does.

The product — two surfaces

Surface	What it is	When it runs
CLI + dashboard	<code>sentinel scan / remediate / dashboard</code>	CI, release gate, audit prep
Bob IDE skill pack	<code>.bob/custom_modes.yaml</code> — 5 custom modes	Every keystroke in your editor

Drop the modes into any team's Bob IDE → `/audit-hipaa` works inline as they code.

Compliance stops being an event. It becomes a habit.

The demo

30 / 100 → 96 / 100. Four minutes. One command.

`sentinel scan` flags 19 HIPAA violations across 9 files (5 PHI data flows, 3 critical findings).

`sentinel remediate --all` opens 19 patch branches with cross-file blast-radius analysis.

Re-scan: green. Audit-ready.

Full 2-minute demo video: [linked in submission](#) · live demo on stage: don't.

Built with Bob IDE — receipts

Bob IDE task	Output	Cost
Audit run via <code>hipaa-auditor</code> mode	JSON findings + auditor notes	\$0.10
Add §164.310 control end-to-end	4 files changed, demo violation seeded	\$1.49
Generate test suite	679 LOC, 30 tests, 30 passing	\$0.39
Refactor dirty-tree no-op	Structured status across CLI + dashboard	\$1.09
Code-review catalog	False-positive / false-negative analysis	\$0.07
Generate per-control docs	26 markdown pages in <code>docs/controls/</code>	\$2.92
	Total: \$6.06 / 40 Bobcoins	

Why this matters to IBM

Compliance is the killer enterprise use case Bob IDE was built for.

Healthcare, fintech, gov: IBM's existing customer base. They can't ship without HIPAA, PCI, FedRAMP.

Sentinel makes Bob IDE the answer to a \$50B/year enterprise spend.

Drop `.bob/custom_modes.yaml` into any project → every developer becomes a compliance-aware coder. No external auditors. No six-month cycles. No spreadsheets.

Thank you.

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Try it: `git clone ... && npm install && npm run demo`